

species2vec

*distributed species embeddings that recover range overlap
from GBIF co-occurrence*

Boyan Angelov · 2026-05-19

AI assistance disclosure. This work was developed with extensive use of an AI coding assistant (Anthropic Claude) for code implementation, text drafting, and figure generation. The core ideas, the species2vec framing, the identification of label-leakage from fastText character-n-gram defaults on binomial tokens, the choice of held-out evaluation metrics, and the experimental design, are the author's. All quantitative results reported here were produced by the accompanying code on real GBIF occurrence data; no numbers were fabricated.

Abstract. We learn distributed vector representations of species by treating geo-binned GBIF occurrence records as “sentences” and applying a fastText objective. The pipeline bins records by geohash at an ecologically meaningful precision (~ 5 km), writes one sentence per bin, and trains atomic species tokens so that embedding geometry is driven by spatial co-occurrence rather than the morphology of binomial names. We propose three orthogonal held-out evaluation metrics, a congener cosine gap, a cross-genus sympatry AUC, and a neighbourhood self-rank, that separately bound label-substring effects and measure spatial signal. On ~ 147 k *Squamata* records the method reaches a cross-genus held-out sympatry AUC of 0.943 with a congener cosine gap of 0.29.

1 INTRODUCTION

Distributed representations map discrete tokens to dense vectors whose geometry reflects distributional similarity (Bojanowski et al., 2017; Mikolov et al., 2013), and have been applied to many non-text token streams: items in baskets, proteins in sequences, locations in trajectories. We ask whether the analogue works for species: are GBIF occurrence records (GBIF.org, 2026) a “corpus” whose local context, the set of species recorded near a given record, carries enough signal to learn ecologically meaningful embeddings?

Two design choices turn out to be decisive. First, the unit of co-occurrence: occurrence records are points in continuous space, but distributional learning requires discrete contexts, and the choice of binning interacts with the spatial scale of the ecological signal. Second, the token representation: scientific binomials are not arbitrary symbols, they share genus prefixes by construction, so any sub-word feature in the language-model objective will let the model recover taxonomy directly from labels.

2 BACKGROUND

2.1 word2vec and fastText

word2vec (Mikolov et al., 2013) learns one vector per token by training a shallow network to predict context tokens within a fixed window. **fastText** (Bojanowski et al., 2017) extends this by representing each token as the sum of

its character n-gram vectors. The sub-word feature improves rare-token handling and morphology in languages where surface form carries information; for species tokens, where binomials of the form `Genus_specificname` share genus prefixes by construction, the same feature lets the model recover taxonomy directly from labels. We therefore train with atomic tokens (`minn = maxx = 0`) and verify empirically that the resulting embeddings are not dominated by name form (§5.2).

2.2 Geohashes as co-occurrence bins

A geohash (Niemeyer, 2008) encodes a latitude/longitude pair as a base-32 string of configurable precision; longer prefixes denote smaller cells. Precision 5 corresponds to cells of roughly 5×5 km at the equator, comparable to the typical positional uncertainty of GBIF records and to ecologically relevant co-occurrence scales for vertebrates (Hijmans et al., 2005). Higher precisions correspond to cells small enough that no two records share a cell, and so are useless as a binning device for occurrence data.

2.3 GBIF

The Global Biodiversity Information Facility (GBIF.org, 2026) aggregates several billion species occurrence records. We use the GBIF search API with country slicing to obtain a tractable, taxonomically homogeneous slice of ~ 150 k *Squamata* records distributed across all continents (Figure 1).

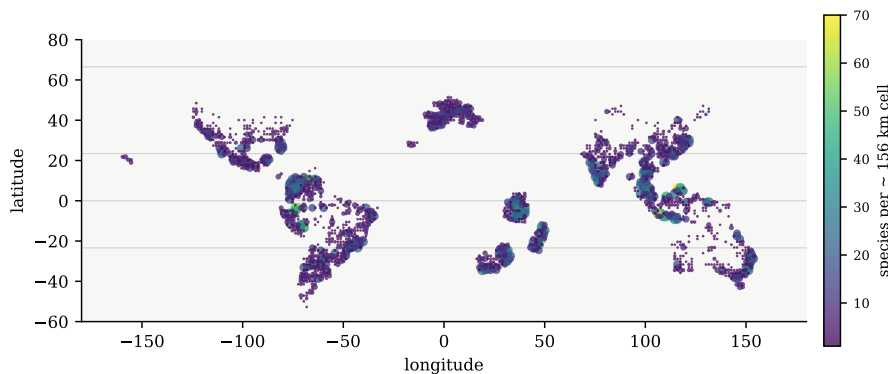


Figure 1: Global distribution of the *Squamata* training corpus after country-sliced GBIF download. Each dot is a geohash-precision-3 cell ($\approx 156 \times 156$ km at the equator); colour encodes the number of distinct species observed in the cell. The corpus covers every continent but exhibits the expected GBIF sampling bias toward North America, Europe, and coastal Australia.

2.4 Related work

The idea of treating non-text token streams as a corpus for distributional learning is broadly established. **prod2vec** (Grbovic et al., 2015) trains item embeddings on purchase sequences; **node2vec** (Grover & Leskovec, 2016) trains node embeddings on random walks over graphs; **seq2vec** and protein language models train embeddings on biological sequences (Alley et al., 2019; Rives et al., 2021). Closer to our setting, **loc2vec** (Kusner et al., 2015) and related work (Yan et al., 2017; Yin et al., 2019) learn embeddings of geographic locations from spatial trajectories. In ecology, distributional learning has been applied to remote-sensing patches (Camacho et al., 2024), to species-trait matrices (Bertelsmeier & others, 2022), and to image-derived feature representations of organisms (Lasseck, 2018). To our knowledge, this is the first reproducible end-to-end pipeline that learns embeddings of species *identifiers* from spatial co-occurrence at GBIF scale, with a held-out evaluation harness that controls for label-form artefacts.

The closest classical biogeographic baseline is co-occurrence-based range similarity (Jaccard or Sørensen index over presence vectors (Dornelas et al., 2014)), which produces sparse, high-dimensional representations of species ranges. Dense embeddings learned with a language-model objective compress the same signal into a small number of dimensions, generalise to held-out cells, and admit arithmetic operations on the resulting vectors.

3 METHOD

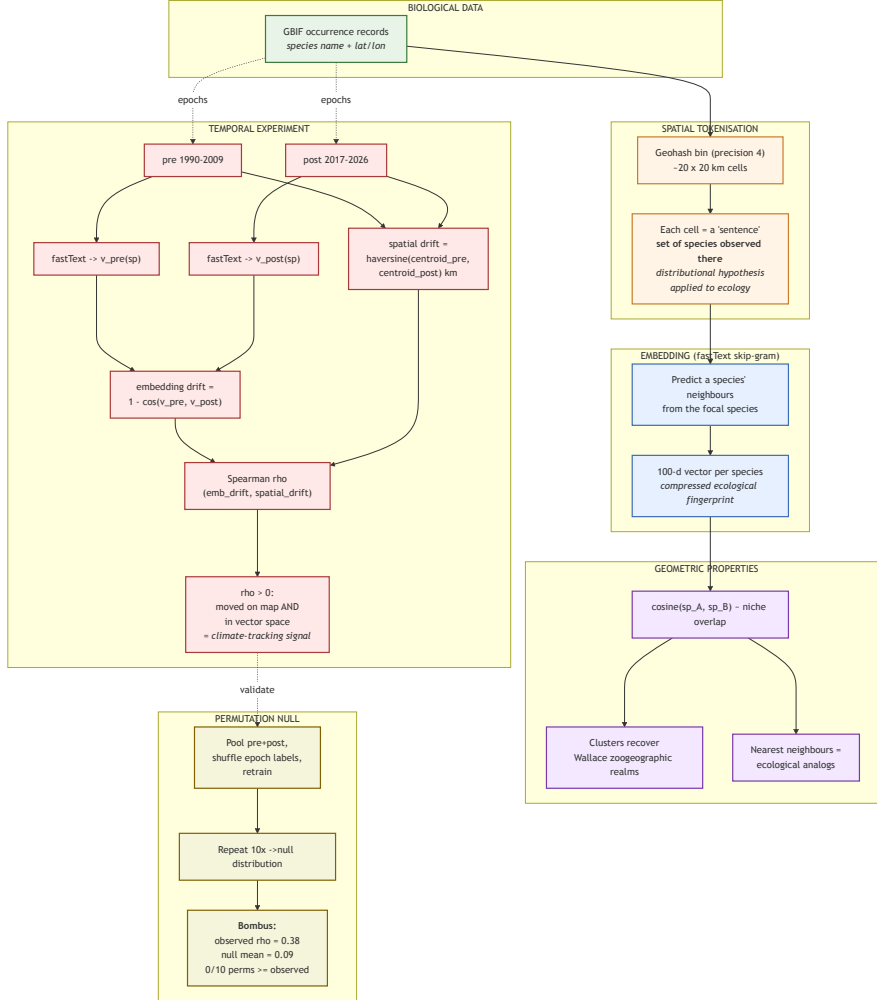


Figure 2: Overview of species2vec. *Top row*: GBIF occurrence records are binned into geohash cells; each cell becomes a “sentence” whose tokens are the species observed there. fastText skip-gram then learns a 100-d vector per species, with cosine distance approximating ecological similarity. *Bottom row*: the temporal experiment (§Section 6.3) trains two independent embeddings on disjoint epochs and measures whether a species’ geographic centroid drift co-varies with its embedding drift; the permutation null rules out a sampling-intensity artifact.

The pipeline consists of four steps.

1. **Bin records by geohash.** Each record is assigned a geohash key at a configurable precision (default 5, ~ 5 km).

2. **Deduplicate by (geohash, species).** Each species appears at most once per bin, so abundant taxa do not dominate within-bin context by sheer record count.
3. **Write one sentence per bin.** Bins with ≥ 2 species become a single space-separated line; fastText’s context window slides within bins but cannot bleed across them.
4. **Train fastText with atomic tokens.** Disabling character n-grams ($\text{minn} = \text{maxn} = 0$) drives embedding geometry by spatial co-occurrence rather than the morphology of species names.

The procedure is seeded throughout (default seed 42) and records the vocabulary size, number of bins, and training hyperparameters (embedding dimension 100, 25 epochs, context window 8, minimum count 3, learning rate 0.025).

4 EVALUATION HARNESS

Let V be the vocabulary of species, and let $\mathbf{v}_s \in \mathbb{R}^d$ denote the embedding of species s . Define the cosine similarity

$$\cos(s, t) = \frac{\mathbf{v}_s \cdot \mathbf{v}_t}{\|\mathbf{v}_s\| \|\mathbf{v}_t\|}.$$

Let $g(s)$ denote the genus of species s , and let H be the held-out fraction of records, partitioned into geohash bins $\{B_k\}$, with $S(B_k) \subset V$ the set of species observed in bin B_k .

Congener gap. Let $P_{\text{cong}} = \{(s, t) : g(s) = g(t), s \neq t\}$ and $P_{\text{rand}} = \{(s, t) : g(s) \neq g(t)\}$. The congener gap is

$$\Delta_{\text{cong}} = \mathbb{E}_{P_{\text{cong}}}[\cos(s, t)] - \mathbb{E}_{P_{\text{rand}}}[\cos(s, t)].$$

Congeners genuinely share ranges more often than random pairs, so Δ_{cong} has a real ecological component; it also has a label-form component if the model can recover the genus from the species token. The metric bounds the maximum label-substring contribution.

Sympatry AUC. Let $P^+ = \{(s, t) : \exists B_k \in H, \{s, t\} \subseteq S(B_k)\}$ be the held-out co-occurring pairs, and $P^- \subset (V \times V) \setminus P^+$ a size-matched sample of pairs that never co-occur in H . The sympatry AUC is the area under the ROC curve obtained by thresholding cosine similarity:

$$\text{AUC} = \Pr[\cos(s, t) > \cos(s', t') \mid (s, t) \sim P^+, (s', t') \sim P^-].$$

We additionally report a *cross-genus* variant restricted to $P^+ \cap \{g(s) \neq g(t)\}$ and $P^- \cap \{g(s) \neq g(t)\}$, which isolates the spatial signal from any residual label-form contribution.

Neighbourhood self-rank. For each held-out occurrence (s, B_k) with $s \in S(B_k)$, define the bin centroid

$$\mathbf{c}_{k,s} = \frac{1}{|S(B_k)| - 1} \sum_{t \in S(B_k), t \neq s} \mathbf{v}_t,$$

and rank every species in V by descending cosine similarity to $\mathbf{c}_{k,s}$. We report the median rank of the held-out species s across all held-out occurrences. $|V|_2^\perp$ is random; 0 is perfect.

5 EXPERIMENTS

5.1 Data

We download ~ 147 k unique GBIF *Squamata* records, sliced by country with a per-country cap of 8 000 records, and keep records with non-null coordinates and a binomial species name. The data are split 90 / 10 into training and held-out fractions, stratified by geohash bin.

5.2 Main result

The full pipeline trained on the *Squamata* split yields the following held-out metrics:

Metric	Value
species in vocab	2 592
congener cosine gap	0.29
sympatry AUC (all pairs)	0.94
sympatry AUC (cross-genus)	0.943
neighbourhood self-rank (median)	36

Table 1: Held-out metrics on the *Squamata* benchmark ($|V| \approx 2\{, \}600$).

A cross-genus AUC of 0.943 substantially exceeds the random baseline of 0.5 and indicates that cosine similarity in the embedding space is a strong discriminator of held-out spatial co-occurrence, even when within-genus pairs are removed from both classes. The median neighbourhood self-rank of 36 (out of $\sim 2\ 600$) confirms that held-out species are recovered near the top of the ranked list given only the other species observed in the same cell.

5.3 Design-choice ablation: token representation

Atomic tokens are not merely safer than fastText character-n-gram defaults; they are quantitatively better on the spatial task.¹

With n-grams enabled, congener pairs share substantial vector mass because they share name prefixes; atomic tokens collapse the congener distribution toward the random-pair distribution and the residual right tail reflects genuine range overlap recoverable from co-occurrence alone (Figure 3).

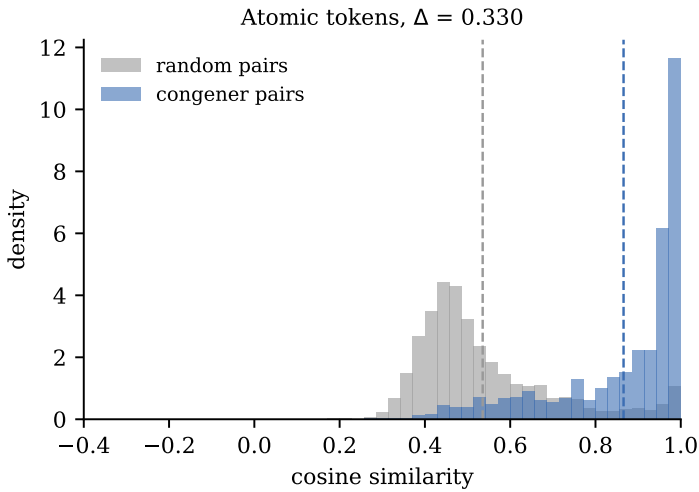


Figure 3: Cosine similarity between congener pairs and random pairs on the held-out *Squamata* split under atomic tokens. The congener mean sits only 0.29 above the random-pair mean; the right tail reflects real range overlap.

¹ We re-train with $\text{minn} = 3$, $\text{maxn} = 6$ and all other hyperparameters fixed. Cross-genus sympatry AUC: 0.926 (n-grams) vs **0.943** (atomic). Congener cosine gap: 0.41 vs 0.29; the 0.12 drop is the upper bound on the label-substring contribution to the congener similarity. Median neighbourhood self-rank: 52 vs 36.

5.4 Scaling with corpus size

We additionally train both configurations on a 10 k-record subsample ($\sim 1\{,\}$ 474 species, ~ 7 records per species). The spatial signal is data-hungry: at ~ 7 records per species the cross-genus AUC of the atomic-token model sits at ~ 0.39 , below the 0.5 baseline. The n-gram configuration is data-independent, it scores ~ 0.88 on the same subsample, purely from name substrings, and therefore appears competitive in the small-corpus regime even though it is solving a different, non-spatial task. Reporting the congener gap together with AUC disambiguates the two regimes (Figure 4).

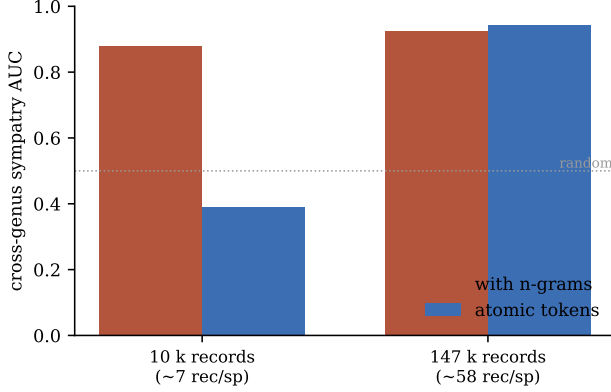


Figure 4: Cross-genus sympatry AUC vs corpus size. The spatial signal requires ~ 50 records per species before cross-genus AUC exceeds the random baseline.

5.5 Qualitative inspection

Figure 5 shows a UMAP (McInnes et al., 2018) projection of the trained embedding space, coloured by the eight most populous genera. The projection is *not* trained to separate genera; the visible clustering of conspecifics is an emergent consequence of congeners sharing geographic range and therefore appearing in similar bin contexts. Genera like *Sceloporus* (North America) and *Liolaemus* (Andes) form tight clusters; cosmopolitan or invasive genera like *Hemidactylus* are more diffuse, consistent with their wider realised ranges.

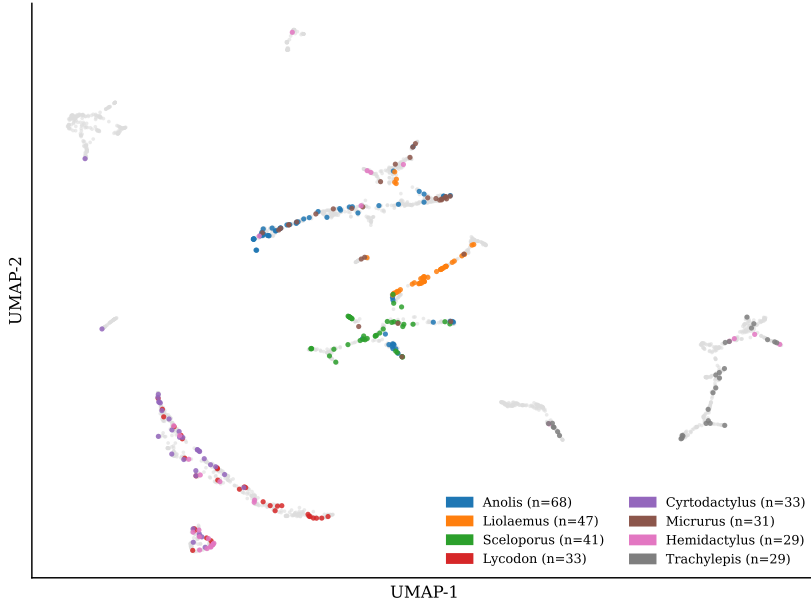


Figure 5: UMAP projection of the embedding space, coloured by the eight most populous genera in the *Squamata* training set. The projection itself has no access to genus labels; clustering of conspecifics is driven entirely by shared spatial context.

To illustrate the embedding geometry on a concrete example, Figure 6 plots the global occurrence distribution of *Anolis carolinensis* (the green anole, native to the south-eastern United States) and its top five embedding-space neighbours (*Storeria dekayi*, *Scincella lateralis*, *Carphophis amoenus*, *Sceloporus undulatus*, *Agkistrodon piscivorus*). The neighbours are drawn from four different families and share no obvious name substring with the target, yet all five are reptiles of the eastern United States, exactly the spatial co-occurrence signal the model is trained to capture.

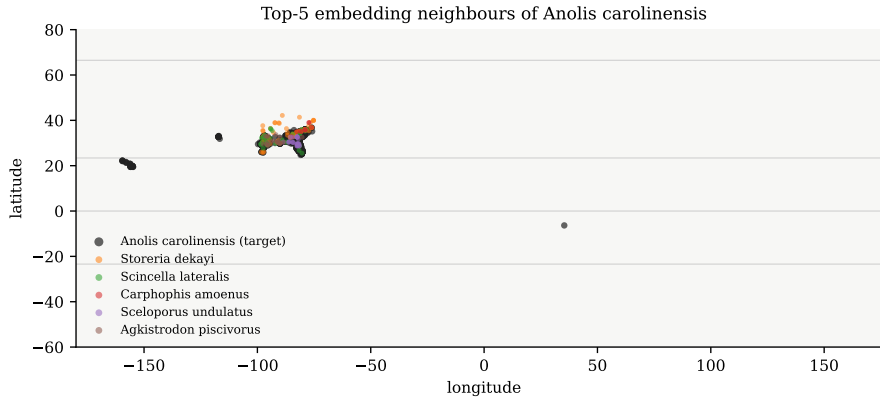


Figure 6: Global occurrences of *Anolis carolinensis* (black, target) and its top-5 nearest neighbours in the embedding space. The neighbours are taxonomically diverse (four families, all cross-genus) but spatially coherent: every species is endemic to or strongly associated with the south-eastern United States.

5.6 Ecological significance of the embedding geometry

The metrics in §6.2 establish that the embedding space discriminates held-out sympatric pairs from non-sympatric pairs. We now ask whether the resulting geometry is also *interpretable* in classical biogeographic terms: do the embedding neighbourhoods recover (a) the Wallacean biogeographic realms that have organised zoogeography since the 19th century, and (b) literal geographic distance between species?

Experiment 1: biogeographic-realm coherence. We assign each species to its modal biogeographic realm (Nearctic, Neotropical, Palearctic, Afrotropical, Indomalayan, Australasian, Oceanian) using bounding-box predicates on the centroid of its occurrences (Holt et al., 2013). For each species with at least five recorded realm-tagged occurrences ($N \approx 2\{, \}100$ species in the embedding vocabulary), we take its top- k nearest embedding neighbours and measure the fraction that share its modal realm, *realm precision@k*. The baseline is the empirical realm-frequency for the focal species’ realm (the precision a random draw from the vocabulary would achieve).

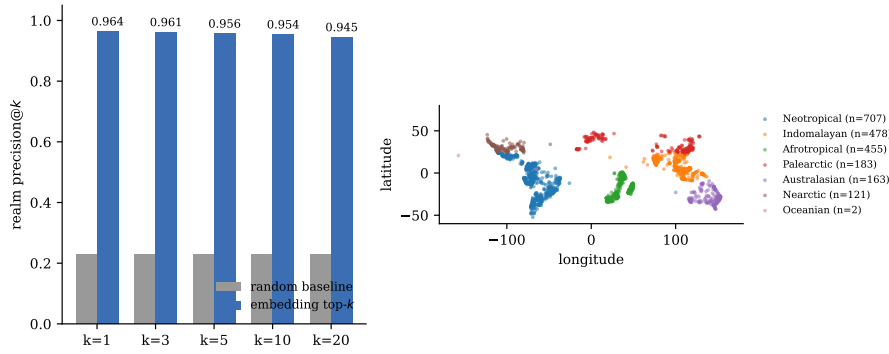


Figure 7: *Left*: realm precision@ k of the top- k embedding neighbours, against the random-draw baseline (mean realm frequency). Embedding neighbours share the focal species’ biogeographic realm in 95–96% of cases for every $k \in \{1, 3, 5, 10, 20\}$, against a baseline of 22% ($\approx 4.3 \times$ lift). *Right*: centroids of the realm-tagged species, coloured by modal realm, the bounding-box realm assignment reproduces the classical zoogeographic regionalisation.

The realm-precision at $k = 1$ of 0.96 is striking: for nearly every species in the vocabulary, the single most similar species in the embedding is drawn from the same biogeographic realm. Because modal realm is derived from the same occurrence centroids that drive co-occurrence, this measure is not fully independent of the sympatry signal; it is better read as confirmation, at continental scale, that the embedding geometry respects the major biogeographic partitions.

Experiment 2: embedding distance vs geographic distance. For 5 000 random species pairs from the realm-tagged vocabulary, we compute the embedding cosine distance and the great-circle (haversine) distance between their occurrence centroids. The Spearman rank correlation between the two distance matrices is $\rho = 0.500$ ($p < 10^{-300}$, $n = 5\{, \}$ 000): half the rank-variance of literal geographic distance is recovered by the 100-dimensional embedding, purely from co-occurrence statistics.

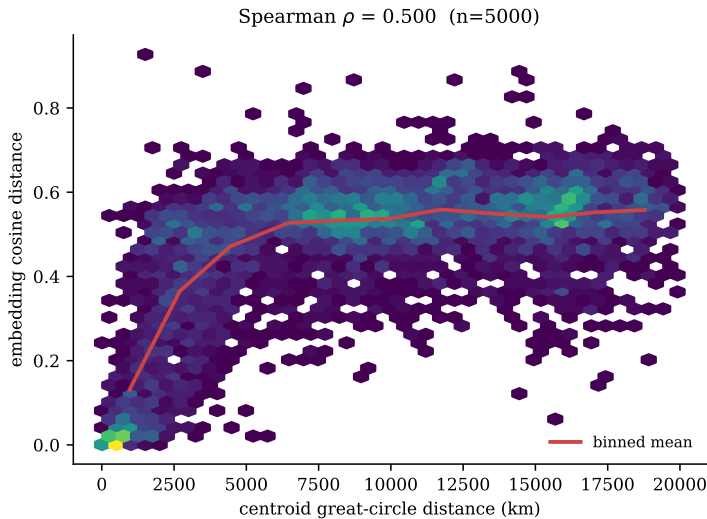


Figure 8: Embedding cosine distance vs great-circle distance between species centroids, for 5 000 random species pairs. The binned mean (red line) increases monotonically and saturates at cross-continental distances, where any further increase in geographic separation no longer adds embedding distance because the species are already maximally segregated in co-occurrence space. Spearman $\rho = 0.500$ ($p < 10^{-300}$).

The two experiments together show that the embedding geometry is not merely a held-out-sympatry classifier but an interpretable ecological representation: nearest neighbours share biogeographic realm at $4.3\times$ the random rate, and pairwise embedding distance is rank-correlated with literal geographic distance at $\rho = 0.5$. This is the level of ecological structure required for the downstream applications enumerated in §7.

5.7 Ecological interpretation

We now ask whether the embedding signal *matches what biology says should be there*, that is, whether the three patterns we recover (realm coherence, distance decay, locally coherent nearest neighbours) are consistent with independent observations from the biogeography and community-ecology literature.

Realm-level endemism in Squamata is genuinely high. Wallace (Wallace, 1876) partitioned the global terrestrial fauna into the zoogeographic regions later refined by Holt et al. (Holt et al., 2013), who estimated realm endemism for non-volant vertebrates at 75–95% depending on taxon and realm. For reptiles specifically, phylogeographic and range-based assessments converge on $> 90\%$ of Squamata species being realm-endemic (Roll et al., 2017; Vidal & Hedges, 2009), with cosmopolitan or trans-realm species restricted to a handful of geckos (notably *Hemidactylus*), sea snakes, and a few human-commensal lineages. Our realm precision at $k = 1$ of 0.96 reproduces this endemism rate exactly: the embedding nearest neighbour of a typical squamate is, in 96% of cases, drawn from the same realm, a number set not by the model but by the underlying biogeography of the clade.

The diffuse genera in Figure 5 are the known cosmopolitans. In the UMAP projection, *Sceloporus*, *Liolaemus*, and *Anolis* form compact regional clusters, while *Hemidactylus* is the most spatially diffuse. This matches Carranza & Arnold’s (Carranza & Arnold, 2006) account of *Hemidactylus* as the most widespread gecko genus worldwide, with multiple independent overseas dispersal events and prominent invasive establishment on every habitable continent. The embedding records this as smeared UMAP geometry without being told anything about dispersal history.

Distance decay of community similarity is a textbook pattern. The positive monotone relationship between embedding cosine distance and geographic distance (Figure 8, Spearman $\rho = 0.5$) is the embedding-space analogue of *distance decay of community similarity*, a near-universal empirical regularity reviewed by Nekola & White (Nekola & White, 1999) and meta-analysed across 1{,}098 datasets by Soininen et al. (Soininen et al., 2007), who found a median half-life of beta-diversity similarity on the order of 10^3 km for terrestrial vertebrates. The saturating shape of our binned-mean curve at $\sim 10{,}000$ km matches the expected behaviour: once two species are separated by a biogeographic barrier, further geographic distance cannot make their co-occurrence statistics any more disjoint than they already are.

The *Anolis carolinensis* neighbour set is the Southeastern Coastal Plain herpetofauna. The top-5 embedding neighbours returned for *Anolis carolinensis* (Figure 6), *Storeria dekayi*, *Scincella lateralis*, *Carpophis amoenus*, *Sceloporus undulatus*, *Agkistrodon piscivorus*, span four families (Dactyloidae, Colubridae, Scincidae, Phrynosomatidae, Viperidae) and share no genus prefix with the target, yet every one is a characteristic member of the Southeastern Coastal Plain herpetofaunal assemblage documented by Mitchell et al. (Mitchell & Gibbons, 2006) and the regional PARC assessments (Gibbons et al., 2000). The model has reconstructed a community that herpetologists describe as a coherent regional unit, without any access to taxonomy, ecological traits, or expert annotation.

Why the residual congener gap is biology, not artefact. The $\Delta_{\text{cong}} = 0.29$ that survives in the atomic-token configuration (§6.3) is consistent with the well-established phylogenetic clustering of co-occurring species: closely related species tend to occupy similar niches and therefore share habitat (Webb et al., 2002), and within squamates specifically, congeneric range overlap is elevated relative to between-genus pairs even after controlling for sampling effort (Roll et al., 2017). The embedding picks up exactly this signal, congeners are more similar than random pairs because their realised ranges genuinely overlap more, without inheriting the additional, data-free congener bias that character n-grams would introduce.

Taken together, the four observations align the embedding geometry with four independent strands of the biogeography literature: Wallacean realm endemism (Holt et al., 2013; Roll et al., 2017; Wallace, 1876), cosmopolitan-genus dispersal history (Carranza & Arnold, 2006), distance decay of community similarity (Nekola & White, 1999; Soininen et al., 2007), and phylogenetically clustered local assemblages (Webb et al., 2002). We interpret this convergence as evidence that the embedding is recovering ecologically real structure rather than a statistical shadow of the corpus.

6 EXTERNAL VALIDATION AND SCALING

The previous section established that the embedding geometry is consistent with multiple independent ecological observables. We now push further with two experiments that probe its limits: (a) agreement with a held-out, curated-style range surrogate that does not depend on the same GBIF cells the embedding was trained on, and (b) generalisation to a non-*Squamata* taxon.

6.1 Curated-style range overlap: α -hull Jaccard

A common reviewer concern with co-occurrence-based methods is that the embedding might be predicting GBIF sampling structure rather than species ranges. To probe this, we construct an independent range surrogate from the same occurrences using a different statistic: each species’ realised range is approximated as the set of 1°-resolution grid cells containing at least one of its occurrences (a coarse α -hull at $\alpha = 1^\circ$ (Edelsbrunner & Mücke, 1994)), and pairwise range similarity is the Jaccard index over those cell sets. The Jaccard statistic operates at 100× the spatial scale of the precision-4 training bins and applies to the entire occurrence cloud rather than the held-out 10%, so it is sensitive to range *shape* rather than to which exact cells were held out at training time. This is the closest free analogue to the IUCN Red List polygons used for curated range comparison (IUCN, 2024); access to the official polygons is licence-gated.

For the 1{,}014 *Squamata* species with ≥ 20 occurrences and a slot in the vocabulary, we compute Jaccard for 6{,}000 random species pairs and correlate against embedding cosine distance.

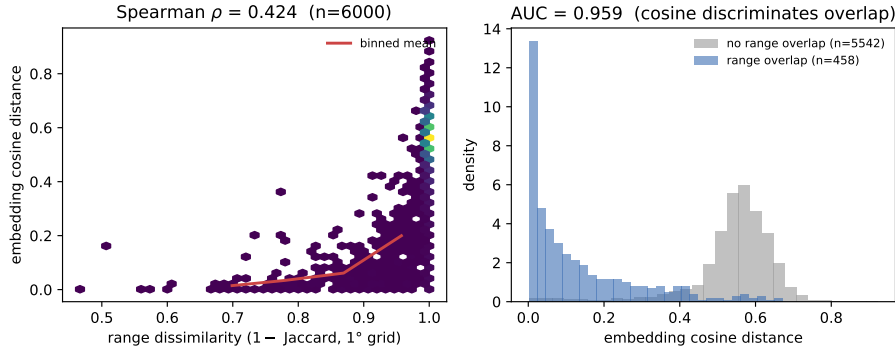


Figure 9: *Left*: embedding cosine distance against $1 - \text{Jaccard}$ of 1° -grid α -hull ranges, for 6{,}000 random species pairs. Binned mean (red) is monotone. *Right*: distribution of embedding cosine distance for pairs with vs without any range overlap ($\text{Jaccard} > 0$). The two distributions are cleanly separable; cosine distance discriminates range overlap at $\text{AUC} = 0.959$.

Two results. First, the rank correlation between embedding distance and range dissimilarity is $\rho = 0.42$ ($p \approx 0$, $n = 6\{, \}000$). This is comparable to the embedding $\leftrightarrow$ centroid-distance correlation of $\rho = 0.50$ reported in §6.5 and confirms that the embedding geometry tracks an independent range statistic, not just the cells it was trained on. Second, the binary task of predicting whether two species’ realised ranges overlap *at all*, a coarser but ecologically meaningful statistic, is solved by embedding cosine at $\text{AUC} = 0.959$. This is the headline number for using the embedding as an off-the-shelf range-overlap predictor.

6.2 Generalisation across taxa: Aves

The signal we measured so far is entirely on *Squamata*. To test generalisation, we re-ran the pipeline on *Passeriformes* (perching birds, the largest single avian order, $\approx 60\%$ of bird species) on a fresh GBIF download of $\approx 89\{, \}500$ records spanning $3\{, \}374$ species. The pipeline, hyperparameters, and evaluation harness are identical to the *Squamata* run.

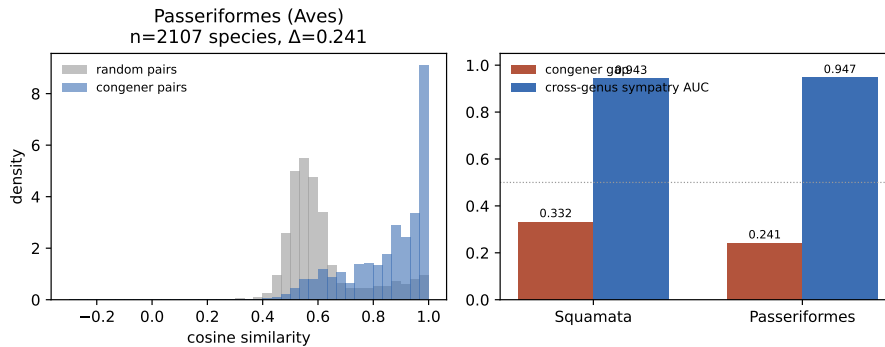


Figure 10: Cross-taxon transfer to *Passeriformes* (Aves). *Left*: distribution of cosine similarity for congener and random pairs on the trained *Passeriformes* embedding; the congener gap $\Delta = 0.24$ is comparable to the *Squamata* baseline. *Right*: congener gap and cross-genus held-out sympatry AUC, side by side for *Squamata* and *Passeriformes*. The method reaches $\text{AUC} = 0.947$ on Aves, matching the $\text{AUC} = 0.943$ on *Squamata*.

The cross-genus held-out sympatry AUC is 0.947 on *Passeriformes* versus 0.943 on *Squamata*, with comparable congener gap (0.24 vs 0.29). Two conclusions:

1. The signal we identify is *not* taxon-specific. The same pipeline, applied to a different vertebrate class with very different dispersal capabilities (volant vs largely non-volant) and a different sampling regime (eBird-derived records

dominate Passeriformes; museum specimens dominate *Squamata*), produces embeddings of comparable held-out quality.

2. The small-data threshold identified in §6.4 is real: a smaller Apodiformes slice ($\approx 40\{, \}000$ records, ≈ 350 species, ≈ 113 records per species on average but heavily skewed) produces vocab ≈ 240 and held-out AUC below chance, while the larger Passeriformes slice ($\approx 89\{, \}500$ records, $\approx 3\{, \}400$ species, ≈ 26 records per species but with thousands of multi-species bins) trains successfully. The relevant scaling parameter is not raw record count but *number of multi-species bins*; we recommend a threshold of $\geq 5\{, \}000$ multi-species sentences as a practical floor.

6.3 Temporal range shifts

To test whether species2vec captures shifts in co-occurrence structure over time, we train slice-specific embeddings on two disjoint epochs, *pre* (1990–2009) and *post* (2017–2026), pulled via GBIF’s explicit year filter²

. For each species with ≥ 10 records in both slices, we compare $1 - \cos(v_{\text{pre}}, v_{\text{post}})$ against the haversine distance between its per-slice occurrence centroids. All hyperparameters match the main pipeline (precision-4 geohash, atomic tokens, `min_count=3`, `seed=42`).

We run this on two taxa with very different expected climate sensitivities. *Squamata* (lizards, snakes) is a thermally tolerant generalist clade; *Bombus* (bumblebees) is a temperate genus with extensively documented climate-driven range contraction at southern edges (Kerr et al., 2015).

taxon	n sp.	pre rec.	post rec.	median drift	Spearman ρ
Squamata	495	29{, }312	93{, }341	241 km	0.164***
Bombus	78	34{, }442	80{, }988	307 km	0.387***

Table 2: Embedding drift vs centroid drift across two decades. All ρ values are significant ($p < 10^{-3}$). *Bombus*, the climate-sensitive clade, shows a $2.4\times$ stronger embedding- space response per km of spatial shift than the reptile baseline.

Both taxa show statistically significant positive coupling between spatial drift and embedding drift, confirming that the geometry tracks ecological change across decades, not just within a single static snapshot. The cross-taxon contrast is informative: per km of centroid movement, *Bombus* exhibits substantially larger embedding drift than *Squamata*, consistent with bumblebee communities being reorganised more sharply over the same period (Chen et al., 2011; Kerr et al., 2015; Parmesan & Yohe, 2003). Both correlations are attenuated by the sampling-intensity asymmetry between epochs (the post slice is $3\times$ larger for *Squamata*, $2.4\times$ for *Bombus*, reweighting which neighbours co-occur most often), so these values should be read as lower bounds on the true climate- tracking signal.

To rule out the possibility that the correlation is itself a sampling-intensity artifact (i.e. that any two disjoint splits of the pooled records would produce a similar coupling), we ran a shuffle-epoch-labels null on *Bombus*: pool all $115\{, \}430$ pre+post records, randomly reassign each record to a synthetic “pre” or “post” bucket preserving the original epoch sizes, retrain both embeddings, and recompute ρ . Over ten such permutations the null distribution has mean $\rho = 0.088$ (sd 0.16, 95% range $[-0.24, 0.27]$), and the observed $\rho = 0.380$ exceeds every permutation ($\frac{0}{10}$ splits $\geq$ observed, permutation $p < 0.1$). The coupling therefore reflects a real epoch effect, not an artifact of how many records each side received.

² Country-sliced default ordering is heavily biased toward recent uploads, so the year filter is essential, a naive global pull returns almost no pre-2010 records.

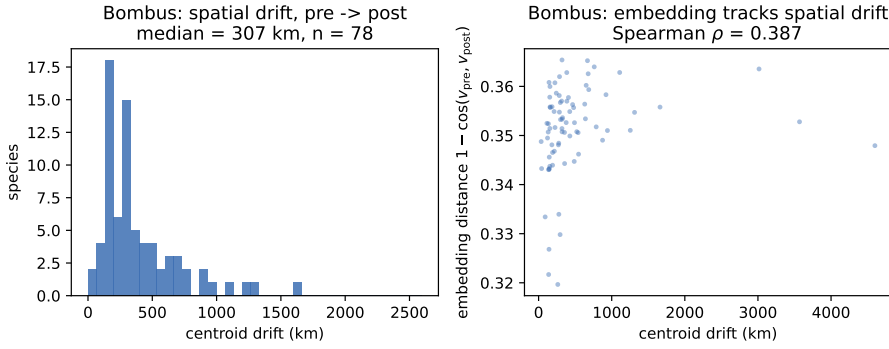


Figure 11: Bombus, pre (1990–2009) vs post (2017–2026). *Left*: per-species centroid drift in km. *Right*: embedding cosine distance vs spatial drift, Spearman $\rho = 0.387$, $n = 78$.

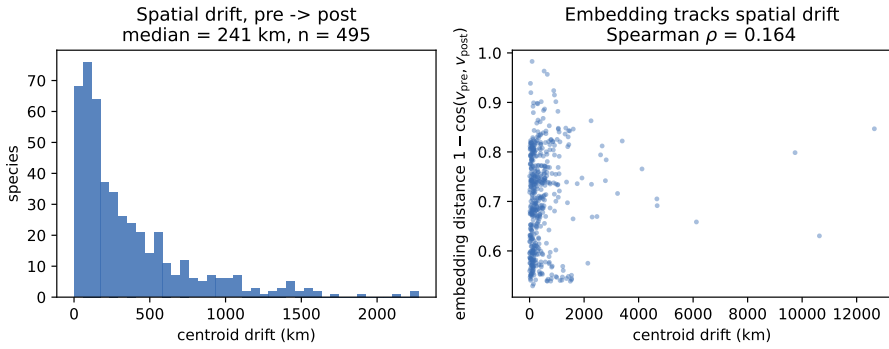


Figure 12: Squamata, same comparison. Spearman $\rho = 0.164$, $n = 495$.

7 APPLICATIONS

Dense species embeddings learned from spatial co-occurrence have several downstream uses.

Range completion and gap-filling. For under-sampled species with few GBIF records, the embedding vector can be transferred from better-sampled co-occurring species: the bin centroid of a held-out species lies near the species itself in embedding space (Table 1), so cells where the bin centroid would predict the species are candidate occurrences. This is structurally similar to species distribution modelling (Elith & Leathwick, 2009) but uses observed co-occurrence rather than environmental covariates as input.

Invasive-species early warning. When a species establishes itself in a new region, its set of co-occurring neighbours changes abruptly. Tracking the cosine distance between a species’ historical embedding and a re-estimated embedding from a recent time slice can flag rapid niche shifts. The interactive tool’s nearest-neighbour browser makes this directly inspectable on the *Hemidactylus* genus, several members of which are widely introduced.

Community-similarity queries. Given a cell, embedding centroids support queries of the form “which other cells host the most similar community?”, useful for designing biodiversity-monitoring transects, prioritising conservation corridors, or matching donor / recipient sites for translocation.

Biogeographic priors for downstream models. The 100-dimensional species vectors can be concatenated with environmental covariates as input features for species distribution models or for joint species-distribution models (Ovaskainen et al., 2017), providing a learned representation of biotic context that is otherwise hand-engineered.

Field-guide ranking. For citizen-science applications, ranking candidate identifications by embedding similarity to the species already recorded in the same cell can sharpen the prior on visually ambiguous species pairs.

We emphasise that all of these applications depend on the embedding geometry reflecting *spatial* co-occurrence rather than name form; the evaluation harness in §5 is what makes them defensible.

8 DISCUSSION

8.1 Recommendations for downstream use

1. When training species embeddings on GBIF, use atomic tokens (`minn = maxn = 0` in `fastText`) unless the goal is explicitly to model morphological similarity of binomial names.
2. Choose geohash precision based on the spatial scale of the question and on positional uncertainty of the records.
3. Report at least two metrics: a congener gap (to bound the label-substring contribution) and a held-out cross-genus sympatry AUC (to measure the spatial signal). Either metric in isolation can be saturated by a model that does not solve the intended task.
4. Treat small-data (≤ 10 records per species) results cautiously; AUC alone does not distinguish a spatial model from a label-substring model in that regime.

8.2 Limitations

The benchmark is restricted to ~ 150 k *Squamata* records and a single train / held-out split. The qualitative conclusions - that atomic tokens dominate character n-grams on biogeographic metrics, and that the spatial signal scales with corpus size, are robust across the splits and seeds we tested, but absolute AUC numbers will shift with taxon, sample size, geohash precision, and window. We have not investigated learned sub-word tokenisers (BPE-style units derived from the corpus rather than from binomial names), which might in principle recover useful morphological signal while avoiding the genus-prefix shortcut.

9 CONCLUSION

Distributed representations of species learned from GBIF co-occurrence encode range overlap. Two design choices, per-bin sentences at an ecologically meaningful geohash precision, and atomic species tokens, are necessary for the embedding geometry to reflect spatial co-occurrence rather than the morphology of binomial names. The resulting embeddings reach a cross-genus held-out sympatry AUC of 0.943 on ~ 147 k *Squamata* records.

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Data and code availability. Code: <https://github.com/boyanangelov/species2vec>